Open/Public Data Catalogue for Sustainable-Finance & Community-Regeneration Hackathons

Research cut-off: 23 September 2026. I have prioritised resources that can be used immediately in prototypes, while also assessing whether they provide a credible route to a production service. UK-highlighted resources are marked 🇬🇧 **UK**.

Best rapid-prototyping stack

OpenStreetMap + WorldPop/GHSL + public socioeconomic data + climate/nature evidence.

Best UK venture stack

ONS/IMD + Companies House + Environment Agency + Natural England + HM Land Registry.

Main commercialisation risk

Open data licences do not guarantee production SLAs; public endpoints often require caching or self-hosting.

Scoring method

Ease score is my assessment, not the provider's:

- **5:** no/instant signup; REST/CSV/JSON; strong examples; hackathon-ready.
- **4:** small authentication step, beta status or moderately complex schema.
- **3:** SDMX, SPARQL, EO formats, asynchronous processing or dataset-by-dataset licences.
- **2:** specialist tooling, approval, fragmented provider endpoints or material operational work.
- **1:** public in principle but impractical for rapid prototyping.

E = emerging team/MVP. **S** = established team/scalable application.

Master catalogue

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
World Bank Sovereign ESG / Indicators API — World Bank; API + downloads; global	Free; Indicators API normally anonymous; [ESG portal]. The portal covers environmental, social and governance indicators and had a January 2026 refresh; reuse terms vary by underlying series. [1]	Authoritative collated national statistics; indicator-specific annual/ periodic updates; inspect source metadata.	Sovereign transition scorecards, natural-capital-linked lending. **E:** country ESG dashboard. **S:** sovereign or DFI screening engine.	**5** — simple country/year structure; excellent metadata.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
UNSD SDG API — UN Statistics Division; REST/CSV/ Excel; global	Free public Swagger API; no authentication shown for read endpoints; [documentation]. [2]	Official SDG data reported via custodian agencies; series expose last-updated metadata; frequency varies.	SDG-linked bonds, community outcome baselines. **E:** local-to-national SDG explorer. **S:** impact-reporting pipeline.	**4** — broad and documented, but multidimensional schema.
Financial Access Survey (FAS) — IMF; API + download; global	Free public dataset; IMF API access route; [dataset/API]. [3]	Annual statistics on financial-service access and usage, including provider/ customer disaggregation.	Financial-inclusion mapping, community-bank benchmarking. **E:** access-gap dashboard. **S:** banking-market-entry intelligence.	**4** — strong metadata; IMF API model needs learning.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
NGFS Climate Scenarios — NGFS/IIASA; REST API + CSV/XLSX; global	Free; Scenario Explorer registration may apply; licence/ disclaimer must be retained; [portal/API]. [4]	Versioned scenario releases, not live data. **Important:** NGFS flags a retraction affecting specified Phase-V physical-loss variables; treat these separately.	Transition/ physical-risk stress testing. **E:** scenario shock calculator. **S:** bank portfolio stress-testing service.	**3** — rich but technically and methodologically demanding.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
ECB climate indicators — ECB; downloadable/ECB statistical interfaces; EU/euro area	Free; no end-user key normally needed; [indicator portal]. [5]	Sustainable-finance indicators are official statistics; carbon-emissions and physical-risk indicators remain analytical and were updated in November 2025.	Sustainable-finance market benchmarking. **E:** EU green-finance dashboard. **S:** regulatory analytics.	**3** — credible but SDMX/ statistical structures.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
Climate TRACE API — Climate TRACE; REST/ OpenAPI; global	Free public API; no key displayed in documented GET example; [v7 API]. [6]	Asset-, city- and area-level emissions; annual source data from 2021 through the current API year. Method/ model caveats matter.	Financed-emissions discovery, industrial transition. **E:** nearby-emitter map. **S:** supply-chain or lending exposure engine.	**5** — modern OpenAPI, filters and code clients.
GLEIF LEI API — GLEIF; REST; global	Free public production API; [documentation/demo]. [7]	Based on GLEIF Golden Copy; entity, ownership, BIC/ISIN mappings.	Counterparty resolution, issuer identity, green-bond entity graphs. **E:** company matcher. **S:** KYC/ESG master-data layer.	**5** — strong search, fuzzy matching and demo.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
UK Companies House API — Companies House; REST/ JSON	Free account/ API key; HTTP Basic using key as username; 600 requests per five minutes; [auth] and [limits]. [8] [9]	Live public register; cache and use bulk products at scale.	SME mapping, retrofit supplier discovery, community-business networks. **E:** local enterprise explorer. **S:** entity-monitoring/ enrichment service.	**4** — excellent value; signup, pagination and limits.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 UK Open Banking Open Data APIs — UK banks/OBL; distributed REST endpoints	Public open-data specifications covering branches, ATMs, accounts, SME loans and cards; no customer consent data; [v2.4 specification]. [10]	Provider-maintained; endpoint coverage and consistency need live testing.	Banking deserts, SME product comparison. **E:** branch/ ATM access map. **S:** community-finance marketplace.	**3** — common schema, but endpoints are distributed.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
Our World in Data Grapher API — OWID; CSV/JSON/ZIP; global	Free, no key; CC BY 4.0 for OWID data packages, but a 403 can identify non-redistributable source data; [API]. [11]	Curated secondary data; metadata supplies source, citation, timespan and update fields.	Fast contextual dashboards across emissions, poverty, energy and food. **E:** pitch-ready evidence. **S:** use upstream primary sources for critical decisions.	**5** — append `.csv` or `.metadata.json`.
HDX HAPI / HDX CKAN API — UN OCHA; API + CSV; humanitarian/global	Public; read access generally anonymous; write token required; HAPI is beta; terms and source licences apply; [documentation]. [12] [13]	Standardises contributor data and exposes warnings/errors; geographic coverage is strongest in humanitarian-response countries.	Resilience, food security, displacement and funding. **E:** vulnerability dashboard. **S:** crisis-finance decision support.	**4** — standardised indicators, but beta/coverage caveats.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
Sarafu/ Grassroots Economics software stack — Grassroots Economics; open API/ software + data schema	Self-hostable FOSS, mostly AGPL-3.0; public developer documentation; not a guaranteed hosted public-data service; [docs]. [14]	Transaction/ event data quality depends on deployment and community consent.	Community asset vouchers, mutual credit, regenerative livelihoods. **E:** sandbox ledger. **S:** locally governed voucher infrastructure.	**2** — highly relevant but specialist deployment and governance.
OpenStreetMap Overpass API — OSM community; query API + planet extracts; global	Free public instances; no key; ODbL attribution/ share-alike obligations. Public instances are for small/ light use; self-host/ extract for production; [guide]. [15] [16]	Community-maintained, frequently changing; quality varies spatially; database lag is normally minutes.	Amenities, transport, vacant/ community assets, accessibility. **E:** asset map. **S:** self-hosted place-data pipeline.	**4** — superb breadth; Overpass QL and operational limits reduce score.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
geoBoundaries gbOpen — William & Mary geoLab; API + GeoJSON/ shapefile; global	Free, anonymous; **gbOpen CC BY 4.0**. Avoid assuming gbHumanitarian/ gbAuthoritative carry identical commercial rights; [API]. [17]	Versioned administrative boundaries with source, licence and build metadata.	Spatial joins and comparable local indicators. **E:** boundary lookup. **S:** global place-data normalisation.	**5** — one predictable endpoint and standard formats.
WorldPop API/STAC — University of Southampton; REST + raster/STAC; global	Anonymous tier: 1,000 requests/day; free key raises limits; [v2 API]. [18]	Modelled high-resolution population/ demographics; API exposes 2015–2030 Global2 data. Treat projections as estimates.	Population exposure, service catchments. **E:** people-at-risk calculator. **S:** infrastructure demand model.	**5** — GeoJSON input, Python/JS/R examples.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
Global Human Settlement Layer (GHSL) — EC JRC/ Copernicus; bulk rasters/ tools; global	Free/open downloads; acknowledgement/citation required; [portal]. [19]	Globally consistent population/ built-up/ settlement series; versioned rather than live.	Urban expansion, disaster exposure, regeneration targeting. **E:** settlement typology map. **S:** long-run urban-risk analytics.	**4** — excellent data, but raster processing required.
🇬🇧 **ONS Beta API** — Office for National Statistics; HTTP/JSON/ CSV	Free, unrestricted, no key; OGL; beta with possible breaking changes; [developer hub]. [20]	Official UK statistics; dataset-specific release cycles and revisions.	Demography, employment, housing, wellbeing. **E:** neighbourhood dashboard. **S:** automated place profiles with version pinning.	**5** — immediate access and good guides; beta caveat.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 **English Indices of Deprivation 2025** — MHCLG; CSV/ XLSX/ geospatial download	Free under public-sector reuse terms; no auth; [official release]. [21]	Accredited official statistics; periodic snapshot, not a live feed. The 2025 release corrected certain lookup files —use the latest versions.	Inclusive investment, grant targeting, social-value baselines. **E:** deprivation-opportunity map. **S:** impact allocation methodology.	**5** — ready-to-join CSV and clear guidance.
🇬🇧 **HM Land Registry Price Paid/ UK HPI** — HMLR; CSV + linked-data/ SPARQL	Free; no key for public linked data; OGL attribution; address-data third-party rights require care; [open-data portal]. [22] [23]	Monthly; transactions available from 1995. Registration lag means newest records are incomplete.	Housing affordability, regeneration additionality. **E:** local affordability map. **S:** property-market monitoring service.	**4** — easy bulk CSV; SPARQL has a learning curve.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 **OS Data Hub OpenData** — Ordnance Survey; APIs + downloads	Free OpenData plan; account/ project key; free and unlimited OpenData usage subject to detail/data limits; [Data Hub]. [24]	Authoritative GB mapping; OS states very frequent source-database updates, though product cadence varies.	Greenspace, roads, buildings and catchments. **E:** authoritative basemap. **S:** GB geospatial platform.	**4** — excellent resources; signup and licensing distinctions.
UK data.gov.uk CKAN API — Government Digital Service; catalogue API	Free; no key or published rate limit; [API docs]. [25]	Metadata quality/ update depends on publisher; catalogue is discovery rather than a uniform data API.	Discover local-authority/ environment datasets. **E:** dataset finder. **S:** metadata harvester with link monitoring.	**5** — frictionless CKAN; actual resources vary.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
GBIF API — Global Biodiversity Information Facility; REST + downloads/ maps; global	Most reads need no auth; account/ basic auth for downloads and writes; source-record licences and citation required; [API]. [26]	Large federated occurrence/ taxonomy corpus; quality and update cadence vary by publisher; GBIF supplies validation and metadata.	Nature baselines, species-risk screening. **E:** species-near-site map. **S:** biodiversity MRV with provenance.	**5** — stable API and R/ Python/Java tooling.
iNaturalist API/datasets — iNaturalist; REST + weekly archive; global	Public reads largely anonymous; OAuth/JWT for protected operations; observation/ photo licences vary; [developer guidance]. [27] [28]	Community observations; research-grade open archive updated weekly, but sampling bias remains.	Citizen-science monitoring, restoration engagement. **E:** nature-observation app. **S:** verification layer combined with formal surveys.	**4** — easy discovery; bulk and licensing discipline needed.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 **NBN Atlas Web Services** — National Biodiversity Network; REST/WMS/ downloads	Public search APIs; login may be required for downloads; each dataset's licence and sensitive-species rules apply; [API]. [29]	UK biodiversity infrastructure, federating partner records; update cadence varies by data partner.	UK biodiversity baseline, planning/ restoration evidence. **E:** species-count map. **S:** due-diligence/ MRV platform.	**4** — rich UK services; SOLR syntax and licence heterogeneity.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 **Natural England Open Data Geoportal** — Natural England; ArcGIS REST/ OGC + downloads	Free/open; mainly OGL, **except where stated**; [portal]. [30]	Definitive English designations/ habitats/ schemes; monthly release process. A peat-restoration layer was temporarily removed in September 2026—do not cache it as current.	Nature recovery, SSSI constraints, stewardship finance. **E:** opportunity/ constraint map. **S:** land-investment evidence service.	**5** — downloadable formats and feature services.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 **EA Flood-monitoring API** — Environment Agency; REST/JSON/CSV	Free, anonymous, OGL; beta/no SLA and not safety-critical; [reference]. [31]	Warnings and feeds refresh up to every 15 minutes, while individual station transfer frequency varies.	Resilience loans, community alerts, property screening. **E:** live flood dashboard. **S:** resilience triage with redundant sources.	**5** — no auth, excellent examples and geofilters.
🇬🇧 **EA Water Quality Explorer API** — Environment Agency; REST + CSV	Public API; authentication not indicated; check dataset-specific rights/ attribution; [API]. [32]	Observations are processed and verified; additions are reflected within two working days and quality-assurance refreshes occur monthly. [33]	River-restoration finance, catchment outcomes. **E:** pollution hotspot map. **S:** outcome-linked funding/MRV service.	**4** — testable docs; new API and large data volumes.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
OpenAQ API — OpenAQ; REST + AWS archive; global	Free account/ API key (`X-API-Key`); source licences must be respected; [v3 API]. [34]	Harmonised ground-level measurements; coverage and sensor/ provider quality vary; hourly/daily/ monthly aggregations exposed.	Clean-air justice and health investment. **E:** exposure map. **S:** community monitoring/ impact analytics.	**4** — modern OpenAPI; key and heterogeneous coverage.
Copernicus Data Space Ecosystem — EU/ESA; STAC, OData, openEO, OGC and downloads; global	Open/free Sentinel access; registration for many downloads/ processing; free processing tier has quotas; [APIs]. [35] [36]	Operational EO archives; dataset-specific revisit and processing levels. Legacy STAC endpoint was deprecated in favour of the current endpoint. [37]	Land-cover change, crops, flood and restoration MRV. **E:** before/after Sentinel map. **S:** scalable EO monitoring service.	**4** — superb tooling; EO expertise and quotas.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
Copernicus Climate Data Store / ERA5 — ECMWF/C3S; API + bulk	Free registration/ token; users must accept each dataset's terms; Python client and notebooks; [CDS API]. [38]	Quality-controlled climate/ reanalysis datasets; update cadence dataset-specific.	Heat, drought and climate-risk baselines. **E:** postcode climate profile. **S:** portfolio physical-risk engine.	**4** — good snippets; large files and climate semantics.
NASA GIBS — NASA Earthdata; WMTS/WMS/ TWMS/ GDAL; global	Free/open; no key for imagery services; acknowledgement requested; [GIBS]. [39]	Over 1,000 imagery products; most updated daily and many available within hours of observation.	Hazard visualisation, fires, floods and land change. **E:** near-real-time map. **S:** operational monitoring visual layer.	**5** — standards-based and browser/GIS friendly.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
ESA CCI Open Data Portal — ESA Climate Office/CEDA; OpenSearch, THREDDS, WMS/WCS/ OPeNDAP	Free/open; largely anonymous; experimental STAC still in development; [interfaces]. [40]	Long-term Essential Climate Variable records; product-specific releases, versioning and scientific documentation.	Climate baselines, adaptation and natural-capital risk. **E:** historic climate layer. **S:** research-grade risk modelling.	**3** — authoritative but specialist formats.
NOAA Climate Data Online v2 — NOAA/ NCEI; REST/ JSON; global stations	Free email token; five requests/ second and 10,000/day; [documentation]. [41]	Quality-controlled station archives; dataset-specific daily/ monthly/ annual products.	Historic hazard and climate normals. **E:** station trend dashboard. **S:** climate-validation layer.	**4** — simple REST; token and station discovery.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
USGS M2M/ EarthExplorer — USGS EROS; REST/ JSON + downloads; global EO	Free account; login/API access; [M2M documentation]. [42]	Access to EarthExplorer inventories including Landsat; archive/ product update varies.	Land restoration and urban-change monitoring. **E:** Landsat scene finder. **S:** automated archive pipeline.	**3** — powerful, but access workflow and imagery processing.
Open-Meteo open data/ API — Open-Meteo; REST + open AWS dataset; global	No key for public endpoint; API data CC BY 4.0, but free hosted use/ commercial terms should be checked; self-hostable AGPL; [licence]. [43] [44]	Forecast/ reanalysis assembled from national services; underlying models and update cycles vary; AWS database is updated hourly.	Weather-sensitive food, energy and resilience apps. **E:** instant location forecast. **S:** self-host or contract for SLA.	**5** — fastest weather MVP route.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
FAOSTAT API — FAO; API + JSON/ CSV/bulk; global	Free; new developer portal/ account/ personal token; [official portal announcement]. [45]	Official food/ agriculture statistics for 245+ countries/ territories; domain-specific periodic releases.	Food security, farm transition, land-use finance. **E:** crop-risk dashboard. **S:** agricultural market/ impact pipeline.	**4** — consistent new interface; login/token.
USDA FoodData Central — USDA; REST + bulk; US/ global branded foods	Free data.gov key; 1,000 requests/ hour/IP; CC0/ public domain; [API guide]. [46]	USDA-maintained composition and branded-food data; product-type update schedules differ.	Nutrition access, local-food procurement. **E:** healthy-food matcher. **S:** procurement /nutrition platform.	**5** — OpenAPI, examples and demo key.

API/dataset; provider; form	Cost, authentication, documentation and licence	Quality and update pattern	Hackathon themes, uses and team applications	Ease
🇬🇧 **Food Hygiene Rating Scheme API** — Food Standards Agency; REST/XML/ JSON; UK	Free, no signup/key; version header required; [API help]. [47]	Same core rating data used by the FSA service; refresh timing depends on local-authority submissions.	Local food economies, venue quality and high-street regeneration. **E:** food-business map. **S:** procurement /compliance service.	**5** — immediate, searchable and geospatial.

How the resources combine

The following architecture shows the most reusable pattern: **join place/community data to environmental evidence, connect it to organisations and finance, then preserve source, licence and date lineage through scoring and monitoring**. [20] [26] [7] [35]

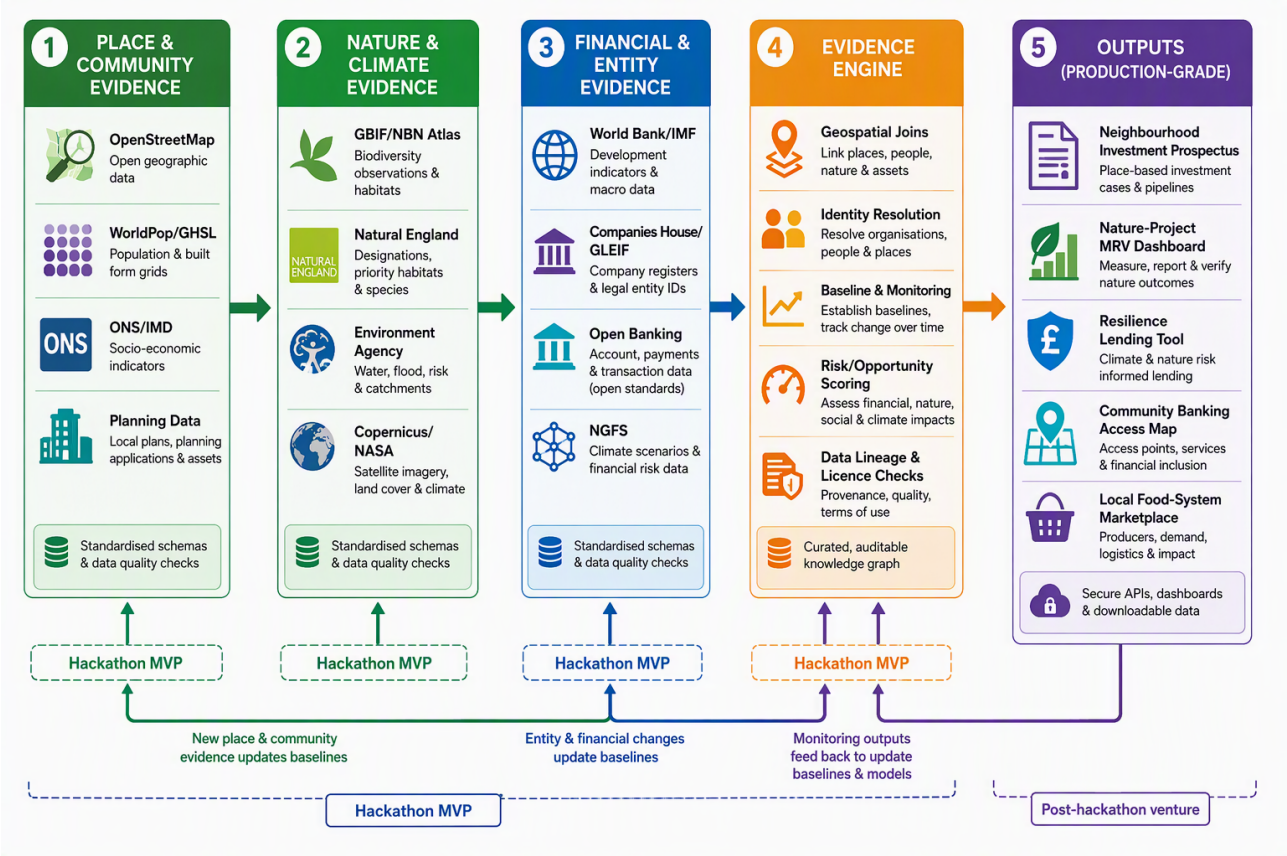


Figure 1: A practical evidence-to-finance stack: open place, nature, climate and financial data are joined into traceable baselines, scores and monitoring services that support both hackathon prototypes and scalable ventures.

Prioritised prototype combinations

1. Neighbourhood Regeneration Investment Navigator — highest practical value

- **Combine:** ONS + IMD25 + OSM + HMLR + Companies House + OS OpenData.
- **Proposition:** rank neighbourhoods by need, enterprise capacity, amenities, housing stress and investable local assets.
- **Users/customers:** councils, community development finance institutions, foundations, regeneration developers.
- **MVP → venture:** interactive map → auditable project pipeline, data subscriptions and investment-readiness reports.

2. Nature-Recovery Finance MRV

- **Combine:** Natural England + NBN Atlas/GBIF + Copernicus Sentinel + EA Water Quality + Companies House/GLEIF.
- **Proposition:** establish ecological baselines, monitor land/water change and identify accountable delivery entities.
- **Customers:** landowners, biodiversity-market operators, lenders, local nature-recovery partnerships.
- **Route:** site dashboard → repeatable monitoring methodology → assurance-quality evidence API. **Do not present opportunistic species observations alone as proof of causality.**

3. Community Climate-Resilience Lending Tool

- **Combine:** EA Flood API + CDS/ERA5 + WorldPop/GHSL + IMD/ONS + HMLR.
- **Proposition:** identify vulnerable people, properties and community facilities, then recommend financeable resilience interventions.
- **Customers:** banks, insurers, councils, housing associations.
- **Route:** flood-exposure map → intervention/cost workflow → lender origination and portfolio-monitoring service. The EA API explicitly has no safety-critical SLA, so production use needs redundancy. [31]

4. Banking-Desert and Community-Finance Opportunity Map

- **Combine:** IMF FAS + UK Open Banking branch/ATM APIs + OSM + WorldPop/ ONS + IMD.
- **Proposition:** show where weak physical access overlaps with deprivation, population demand and absent SME products.
- **Customers:** credit unions, mutuals, banks, regulators and local authorities.
- **Route:** access-gap visualisation → partnership/branch/mobile-service planner → recurring market intelligence.

5. Local Food-System Finance Marketplace

- **Combine:** FAOSTAT + FSA hygiene ratings + OSM + Companies House + Open-Meteo + USDA FoodData Central.
- **Proposition:** connect local producers, institutions and food businesses while adding weather, nutrition and operating-quality signals.

- **Customers:** councils, schools/hospitals, wholesalers, food hubs and community lenders.
- **Route:** supplier map → procurement marketplace → working-capital and supply-resilience product.

6. Industrial Transition and Place-Based Just-Transition Monitor

- **Combine:** Climate TRACE + Companies House/GLEIF + OSM + World Bank/ONS socioeconomic indicators + Copernicus.
- **Proposition:** connect emitting assets to owners, workers, communities and observable transition activity.
- **Customers:** lenders, local economic partnerships, unions, transition-finance teams.
- **Route:** asset/emissions map → engagement workflow → financed-emissions and transition-plan monitoring.

7. Community Asset Voucher / Regenerative Economy Sandbox

- **Combine:** Sarafu FOSS + OSM + local food/business data + OWID/ONS indicators.
- **Proposition:** model locally redeemable commitments for food, care, repair or restoration work.
- **Customers:** community organisations, cooperatives and municipal innovation teams.
- **Route:** closed sandbox → consent-led pilot → governed local platform. Legal, safeguarding, payments, privacy and financial-promotion advice would be essential before live deployment.

Important gaps and cautions

1. **Corporate sustainability disclosures remain fragmented.** There is still no single genuinely open, global, company-level API equivalent to commercial ESG terminals.
2. **Nature data records observation, not automatically ecological additionality.** Credible nature-finance MRV needs baselines, controls, field methods and uncertainty disclosure.

3. **Open endpoints rarely guarantee SLAs.** Cache responses, retain provenance and dates, and migrate heavy workloads to bulk downloads, cloud copies or self-hosting.
4. **Licences propagate through combinations.** OSM's ODbL, dataset-specific biodiversity licences, imagery terms and address-data rights require a machine-readable licence register.
5. **Scoring should remain explainable.** Keep raw indicators, transformations, model versions and missing-data flags available to users rather than producing an opaque "regeneration score."